PURCHASE DESCRIPTION FOR THE ACQUISITION OF ONE (1) COMPUTER NUMERICALLY CONTROLLED (CNC) PUNCH/FIBER LASER COMBINATION MACHINE

1.0 SCOPE.

1.1 Scope. The content of this specification describes the minimum requirements for the acquisition of one (1) Computer Numerically Controlled (CNC) Punch/Fiber Laser Combination Machine. This specification also includes the minimum requirements for technical data, tooling, accessories, additional ancillary equipment, foundation, delivery, unloading, rigging, installation, testing, and training. Delivery, installation, and training will be at Fleet Readiness Center East (FRCE) aboard Marine Corps Air Station (MCAS) Cherry Point, North Carolina. The machine will be used to process aircraft sheet metal engine components in Shop 93213, Building 137 for all air systems at FRCE. The machine will primarily be used for precision punching and laser cutting operations. Materials processed include various types and tempers of aluminum, steel, inconel, titanium, copper, brass, nickel and stainless-steel alloys. The Contractor shall be responsible for furnishing, installing, and integrating all components as described within this specification.

2.0 APPLICABLE DOCUMENTS. The documents listed in this section are cited in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed in section 2. Unless otherwise specified, the documents listed in the solicitation or contract shall be the current edition at time of solicitation.

2.1 Government Documents. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the documents listed in the solicitation or contract shall be the current edition at time of solicitation.

2.1.1 Specifications, Standards, and Handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein.

STANDARDS – FEDERAL

FED-STD-H28B: Screw Thread Standards for Federal Service

U.S Department of Labor, Occupational Safety and Health Administration (OSHA)

29 CFR 1910: General Industry, OSHA Safety and Health Standards

(Application for copies should be addressed to the OSHA Publications Superintendent of Documents P.O. Box 37535 Washington, D.C. 20013-7535)

MILITARY STANDARDS

MIL-STD-130N w/Change 1: Identification Marking of U.S. Military Property Item Unique Identification Marking (IUID)

(Application for copies should be addressed to Air Force Product Data Systems Modernization Standards (TMSS) Office, 554 ELSG/SBP, 4375 Chidlaw Road, Bldg. 262, Room S008, Wright-Patterson AFB, OH 45433-5006, Phone (937) 257-0870 or 0853 or https://quicksearch.dla.mil/qsSearch.aspx)

2.1.2 Other Government Documents, Drawings, and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the documents listed in the solicitation or contract shall be the current edition at time of solicitation.

LOCAL FACILITY DOCUMENTS

RESERVED

2.2 Non-Government Publications, Specifications, Standards, and Handbooks. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the documents listed in the solicitation or contract shall be the current edition at time of solicitation.

American Gear Manufacturers Association (AGMA)

AGMA ISO 1328-1-B14: Cylindrical Gears - ISO System of Flank Tolerance Classification - Part 1: Definitions and Allowable Values of Deviations Relevant to Flanks of Gear Teeth

(Address application for copies to: 500 Montgomery Street, Suite 350, Alexandria, VA 22314-1581 USA)

American National Standards Institute, Inc. (ANSI)

B11/STD B11.18: Safety Requirements for Machines Processing or Slitting Coiled or Non-Coiled Metal

B11/STD B11.21: Safety Requirements for Machine Tools Using Lasers for Processing Materials

ANSI Z136.1/Z136.7/Z136.9: Safe Use of Lasers

ANSI/ASHRAE 52.2:Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

(Address application for copies to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036)

American Society of Mechanical Engineers (ASME)

ASME Y14.100: Engineering Drawing Practices

(Applications for copies should be addressed to ASME International Three Park Avenue New York, NY 10016-5990 or www.asme.org)

International Organization for Standardization (ISO)

ISO 4413: Hydraulic fluid power-General rules and safety requirements for systems and their components

ISO 4414: Pneumatic Power-General rules and safety requirements for systems and their components

ISO 9013: Test Conditions for Thermal cutting Geometrical products specification and quality tolerances

(Application for copies should be addressed to the International Organization for Standardization (ISO) 1 rue de Varembé, Casepostale56, CH211Geneva 20, Switzerland)

National Electrical Manufacturers Association (NEMA)

NEMA 250: Enclosures for Electrical Equipment (1000 V Maximum)

NEMA ICS 1: Industrial Control and Systems General Requirements

NEMA ICS 2: Industrial Control and Systems Controllers: Contactors and Overload Relays

NEMA MG 1: Motors and Generators

NEMA Z535.4: Product Safety Signs and Labels

(Application for copies should be addressed to the National Electrical Manufacturers Association, 1300 N 17th Street, Rosslyn, VA 22209)

National Fire Protection Association (NFPA)

NFPA 70: National Electrical Code

NFPA 79: Electrical Standard for Industrial Machinery

2.3 **Order of Precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulation unless a specific exemption has been obtained.

2.4 **Acronym Definitions Not Explicitly Listed in Text.** The following acronyms not explicitly listed in each section text are defined as follows:

a. JIS- Japanese Industrial Standards
b. DIN- German Institute for Standardization
c. EN- European Standard

2.5 **Proposal Format.** Proposal shall be submitted using the following mathematical conventions:

a. Whole numbers will be expressed where one thousand is written as 1,000.
b. Decimals will be expressed where one-tenth is written as 0.1.

3.0 REQUIREMENTS.

3.1 **Design.** All components provided shall be new and current commercially available. Products such as a prototype, pre-production, or experimental do not qualify as meeting the requirements specified herein. The work shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage or distortion shall be accessible for adjustment, replacement, and repair. All parts subject to wear, corrosion, rust, erosion or cavitation shall be constructed of materials that provide the specified performance and reliability.

3.1.1 **Measuring and Indicating Device Graduations.** All measuring and indicating devices on the machine shall be graduated in the United States customary units. An electronic control system that demonstrates switching between manufacturer standard units to United States customary units will be found acceptable.

3.1.2 **Controls.** All electrical, mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's workstation(s).

3.1.3 **Safety and Health Requirements.** All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910. Safety signs and labels on all equipment shall comply with NEMA Z535.4 and ANSI Z136.1/7/9.

3.1.3.1 **Safety Guarding.** Covers, guards, or other safety devices shall be provided per ANSI B11.18/21 or equivalent ISO/EN/DIN/JIS standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be easily removed to facilitate inspection, maintenance, and repair of the parts. Light curtains and fence protection for the machine shall be provided and installed.

3.1.4 Audible Noise Level. Audible noise emitted by the equipment shall not exceed 84 decibels (dB), measured on the “A” weighted scale of a standard Type II sound level meter, at the operator's work position or any point at a distance of three (3) feet from the equipment. Noise generated by the work piece shall be excluded in determining compliance of the equipment with the 84 dB requirements.

3.1.5 Hazardous Material Exclusions. Supplies or materials being provided as part of the equipment shall be free of known hazardous materials. These materials shall not be brought on site without prior approval of the Government. Radioactive materials or instruments capable of producing ionizing radiation as well as materials which contain asbestos, mercury, lithium, methylene chloride, lead (= or >0.06%), or polychlorinated biphenyls (PCB’s) are prohibited. Definitions of hazardous materials are specified in the latest version, including revisions adopted during the term of the contract, of FED-STD-313F. Nickel Metal Hydride and lithium batteries are permissible for memory backup. Class I Ozone Depleting Substances as defined in 40 CFR 82 shall not be used in the performance of this contract or be provided as part of the equipment. Exceptions to the prohibition of these materials must be referred to the Contracting Officer in writing, for consideration, not later than 60 calendar days after effective date of contract and prior to any work being performed.

3.1.6 Environmental Protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.1.7 Lubrication. Means shall be provided to ensure proper lubrication for all moving parts in accordance with manufacturer requirements. All oil holes, grease fittings, and filler caps shall be easily accessible.

3.1.7.1 Lubrication Data. The Contractor shall provide the following information on all liquid or grease lubricants contained within equipment/system or in bulk (5 gallons or more) supplied under this procurement.

a. Liquid Lubricants
    i. Viscosity grade in ISO units
    ii. Viscosity Index
    iii. AGMA and/or SAE classification as applicable
    iv. Mil Spec and/or NATO Code as applicable
    v. Viscosity in Centistokes (cst) @ 40° C and 100° C
    vi. Type Additive package. Example: ZDDP, Rust inhibitors, Foam inhibitors.
    vii. Neutralization, the TBN (alkalinity) and/or TAN (acidity).
    viii. Flash point
    ix. Pour point
b. Grease Lubricants

i. National Lubrication and Grease Institute (NLGI) Number
ii. Type and percent of thickener
iii. Dropping point
iv. Maximum Service Temperature
v. Base oil viscosity range in SUS or centipoise

3.1.7.2 **Recirculating Lubrication.** When the machine is equipped with a recirculating lubrication system, it shall include a filter that is cleanable or replaceable. Each lubricant reservoir shall have at least a 24-hour capacity. Means shall be provided to indicate a low lubrication condition. When a low lubrication condition occurs, the machine shall complete the current operation and come to a complete stop. The machine shall indicate that lubrication is required.

3.1.8 **Replacement Parts.** All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts shall be interchangeable without requiring modification. Replacement parts are those that are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign-sourced replacement parts shall either be interchangeable with domestically supplied components or be readily available from a domestic source.

3.2 **Construction.** The medium shall be constructed of parts that are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without distortion.

3.2.1 **Castings and Forgings.** All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, filling with solder or paste shall be used for reclaiming any defective part.

3.2.2 **Fastening Devices.** All screws, pins, bolts, and other fasteners shall be installed to prevent unintentional loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.2.3 **Threads.** All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28B and the applicable "Detailed Standard" section referenced therein or equivalent ISO/EN/DIN/JIS standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.2.4 **Hydraulic System.** Under the condition that a machine has a hydraulic system, the system shall conform to ISO 4413. The system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. A machine that does not require a hydraulic system to operate properly shall be considered acceptable.

3.2.5 **Pneumatic System.** Under the condition that a machine has a pneumatic system, the system shall conform to ISO 4414. The system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. A machine that does not require a pneumatic system to operate properly shall be considered acceptable.

3.2.6 Gears. All gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in drive trains shall be hardened and ground steel in accordance with ISO 1328-1-B14.

3.2.7 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. The condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.8 Welding, Brazing, or Soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.2.9 Painting. All new equipment shall be painted in accordance with the manufacturer's commercial practice. All paint shall be of the lead-free type.

3.2.10 Workmanship. Workmanship of the machine shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this purchase description.

3.3 Components. The machine shall consist of not less than the following components:

3.3.1 Punching Turret Station. The number of punching stations and indexable punching stations shall be in accordance with Table I. The punching station shall include a minimum of three (3) multi tools and the turret shall contain multi tool functionality. All turret stations shall be sleeved for ease of future repair.

3.3.2 Base. The base shall be a stress relieved, heavy duty, ribbed casting or steel weldment having characteristics that minimize both thermal and vibration effects, while providing the strength and size to assure maximum rigidity and support for any load within the rated capacity of the machine. Means shall be provided for accurate, ease of leveling.

3.3.3 Table. The table shall be a combination of brushes and roller bearings around the punching station and shall ensure part material remains scratch free during processing.

3.3.4 Table Grid. The worktable grid for both laser cutting and punching shall encompass the entire Nominal Worksheet Size outlined in Table I. Roller and brush combination type auxiliary extension table supports shall be provided to support the entire sheet of material. Sheet material extending past the table surface shall be deemed unacceptable.

3.3.5 Table Part Chute. The table in 3.3.3 shall include a part chute for ejecting completed parts. The part chute shall include a sensor and be activated and operated by the control and or programming software provided in 3.3.21. The rectangle shaped chute size opening shall meet the requirement outlined in Table I.

3.3.6 **Work Holding Clamps.** Three (3) automated clamps for work holding shall be provided. Work clamps shall include repositioning capability. A means of piece part/sheet pullout detection shall be provided.

3.3.7 **Foot Switch.** A foot switch shall be provided with equipment for the operation of the clamps during loading and unloading.

3.3.8 **Laser Scrap Conveyor.** A suitable scrap conveyor shall be supplied with the machine. All components to make the conveyor a complete and usable system shall be supplied.

3.3.9 **Scrap Hopper for Laser Scrap Conveyor.** A scrap hopper shall be provided and fit comfortably without modification under conveyor.

3.3.10 **Punch Slug Conveyor.** A suitable scrap punching conveyor shall be supplied. All components to make the conveyor a complete and usable system shall be supplied.

3.3.11 **Scrap Hopper for Punch Slug Conveyor.** A scrap hopper shall be provided and fit comfortably without modification under conveyor.

3.3.12 **Parts Conveyor.** A parts conveyor belt with collection pan shall be provided with machine. A bin for collecting parts from part chute and conveyor shall be provided.

3.3.13 **Nozzle Cleaner.** An integrated nozzle cleaner to automatically remove debris, dust, and spatter from the nozzle shall be provided.

3.3.14 **Drive System.** The drive system provided on the machine shall be a rack and pinion style and AC servo motor driven, utilizing ball screws and links. A combination of rack and pinion or ball screw drives are acceptable.

3.3.15 **High Speed Capacitive Cutting Head.** The High-Speed Capacitive Cutting Head shall be a non-contact cutting head suitable for a variety of cutting applications, including piercing of all materials and thicknesses listed in Table I, and equipped for following distorted material. The High-Speed Cutting Head shall include the following features, as a minimum:

a) Collision Detection System
b) Protection Window
c) Spare Protection Window Holder
d) Spare Protection Window

3.3.16 **Dust Collector.** The dust collector shall be integrated into the Fiber Laser Cutting System such that it will remove all smoke and dust generated by the cutting operation, filter the air, and return the filtered air to the inside shop atmosphere. Filter efficiency shall meet the requirements of ANSI/ASHRAE 52.2. Filter efficiency test results shall be provided with the dust collector. The dust collector shall be located inside the building and contain no less than the following features:

a) Particulate emissions of no more than .05 grams per dry, standard cubic foot
b) Continuous duty, pulse jet cleaned, cartridge style High Efficiency Particle Air (HEPA) filters
c) Filter efficiency of 99.99% at 1.0 micron
d) Solenoid operated diaphragm valves shall pulse sequentially
e) Inlet of pneumatic system shall demonstrate the ability of accepting 85 psig compressed air
f) Compressed air inlet shall be at a single connection point.
g) Differential pressure gauge for monitoring condition of filters
h) Spark Arrestor

3.3.17 **Chiller Unit.** When a chiller unit is required by the manufacturer to keep the laser resonator and optics at a constant temperature, the chiller unit shall be provided and installed inside the building and include all materials needed to connect the chiller to the laser cutting machine. British Thermal Unit (BTU) capacity shall be determined by the vendor for the machine provided.

3.3.18 **Gas and Assist Gas System for Laser Cutting Acceptance.** The following gas shall be provided for destination acceptance in 4.5. Refer to designated locations of gas supply tanks on Attachment D drawing PE-22170.

3.3.19 **Auxiliary Oil Cooler Unit.** An auxiliary oil cooling unit shall be provided with the machine

3.3.20 **Nitrogen Gas.** Bottled Nitrogen Gas shall be provided for the operation of machine during both origin and destination acceptance. The quantity of tanks supplied shall be sufficient for the duration of the final acceptance inspection. The Nitrogen tanks shall be properly secured inside building 137 near the machine location of equipment installed.

3.3.21 **Oxygen Assist Gas.** Bottled Oxygen Gas shall be provided for the operation of machine during both origin and destination acceptance. The quantity of tanks supplied shall be sufficient for the duration of the acceptance. The Oxygen tanks shall be properly secured inside building 137 near the machine location of equipment installed.

3.3.22 **Control System Requirements.** The Programmable Logic Controller (PLC) hardware shall be preserved on the control panel with the exception of the controller module. The controller module shall be upgraded to the latest compatible hardware version in accordance with NEMA ICS 1/2.

3.3.22.1 **Cam Software.** A licensed Computer Aided Manufacturing (Cam) Software solution shall be provided on the required Laptop 3.3.22.2 with the capability to generate laser cut and punching toolpaths for all available functionality, material types, and material thicknesses of the machine's capability and outlined in Table I. The post processor for the machine shall be provided by the same OEM as the software. The software provided shall be used for the punching/laser cut performance acceptance test in section 4.7.3. The software solution shall contain standard computer aided design (CAD) functions and custom functionality including but not limited to the following:

a) Nesting
b) Punching
c) Laser Cutting
d) Nibbling
e) Forming
f) Countersinking
g) Wheeling
h) Scribing
i) Marking
j) Deburring
k) Engraving
l) Tapping
m) Cutting tables for all available material types and tempers including but not limited to Aluminum, Steel, Inconel, Titanium, Copper, Brass, Nickel, and Stainless-Steel alloys
n) DXF import and export capability

3.3.22.2 **<u>Laptop for Offline Programming.</u>** A laptop shall be provided with the following characteristics:

a. Dell Pro Max 18 Plus
b. Operating System: Windows 11 Pro English
c. Processor: Intel Core Ultra 9 285HX (13 TOPS NPU, 24 Cores, 24 Threads, up to 5.5 GHz, 55W)
d. Memory: 64GB, 2x32GB, DDR5, 6400 MT/s, CSoDIMM, Non-ECC
e. Video Graphics Card: NVIDIA RTX PRO 5000 Blackwell, 24GB GDDR7
f. Hard Drive: 1TB, SSD, Gen 5, SED
g. Optical Readers shall be external
h. Intel Wi-Fi 7 BE200, 2x2, 802.11be, MU-MIMO, Bluetooth 5.4 wireless card
i. Wireless Driver: Intel Wi-Fi 6E AX211 160MHz
j. Ports and Slots:
   i. One (1) Audio Jack port
   ii. One (1) Thunderbolt 4 , Type C
   iii. Two (2) USB A
   iv. One (1) RJ45
   v. One (1) HDMI
   vi. Two (2) Thunderbolt 5, USB Type C
   vii. One (1) Sd Card Reader
   viii. One (1) Smart Card Reader.
k. TPM Module (Trusted Platform Module 2.0)
l. 110 Volt Power Cord
m. 1 Year onsite service with 1 Year NBD Limited Onsite Service After Remote Diagnosis
n. Palm Rest, with Smart card only (Microsoft Usbccid Smartcard Reader (WUDF))

o. 18in Anti-Glare Laptop Screen with Camera and Mic
p. CAT 6 cable connection to equipment, computer, and facility local area network, all connections simultaneously active.
q. Shall be TAA compliant.
r. MAC address for laptop shall be provided to customer at least 60 days prior to delivery.

3.3.22.3 Accessories for Laptop. The following laptop accessories shall be provided:

a. Two (2) Dell Ultra Sharp 24in Infinity Edge Monitors - U2422H
b. Dell MDS19 Dual Monitor Stand
c. 240W AC Adapter
d. Dell Smartcard Keyboard KB813/ with corded mouse.
e. Dell Performance Dock - WD19DC 240W PD

3.3.22.4 Maintenance Laptop with Troubleshooting Software. An engineering maintenance laptop with a permanently licensed software seat loaded onto the laptop shall be provided with the machine. The laptop and software shall demonstrate troubleshooting of the PLC, Controls, and Drives. The laptop computer provided shall be sufficient to run the software without error. Use of the software shall be included in maintenance training called out on 3.10.3

3.3.23 Punch Spray Lubrication. Punch spray lubrication shall be provided to increase tool life and shall be programmable using different intensity levels.

3.3.24 Warped Sheet Detection. An automatic stop of processing shall be supplied for when warping of sheet is detected.

3.3.25 Electrical System. The machine shall conform to NFPA 79 or current equivalent ISO/EN/DIN/JIS/IEC standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements. The existing source power available at the facility is 460 volt, 3-phase, 60 Hertz (Hz), 250 Amps available. The machine's electrical system shall be tolerant of fluctuation of ± 10 percent. The electrical system shall be complete, including any electrical transformer(s) that may be required to modify the existing source voltage to the proper operating voltage of the equipment. A properly rated, and fused, single disconnect device shall be utilized on the machine with means of lockout in the off position only.

3.3.25.1 Electrical Enclosures. Construction of the electrical enclosures shall be in accordance with NEMA 250 type 12 or current equivalent ISO/EN/DIN/JIS/IEC standards and shall provide for ventilation of components and shall be designed for an indoor non-hazardous location. All electrical components shall conform to NFPA 79 or equivalent ISO/EN/DIN/JIS. The machine shall be provided with interlocks on its electronic cabinet that shall instantly remove power from the equipment when the door is opened in accordance with NFPA 79 or current equivalent ISO/EN/DIN/JIS/IEC standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.3.25.2 **Arc Flash Protection.** The machine enclosures shall be marked for arc flash in accordance with NFPA 79 Paragraph 16.2.3 "Safety Signs for Electrical Enclosures" or current equivalent ISO/EN/DIN/JIS/IEC standards for arc flash safety. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements.

3.3.26 **Controller Features.** The machine shall be provided with a CNC controller, incorporating all standard features. The CNC unit shall provide programmable automatic control of machine functions, operating modes, axis movement and other program directed functions for the machine. Movement in all axes shall be independent and shall be capable of independent positioning. Control features shall include Manual Data Input (MDI), simultaneous control of all axes, programmable interface, part program and buffer edit capability, part program storage, fixed cycles, and control diagnostics to at least board level. The CNC shall direct machine functions from command data stored in memory. The control shall initiate a halt and error signal should a fault condition occur in the control. The control shall automatically shut down when internal operating temperatures exceed safe operating limits. MAC addresses for laser/punch shall be provided at least 30 days prior to delivery. The controller shall also contain the following minimum features:

a) 19-inch touch screen
b) Ability to limit control functions to match user level
c) Parts and Nest Thumbnail views
d) Graphic Layout for tool installation
e) Display of material type, tooling, and clamp setup
f) Machine setup information
g) Engraving Function
h) Adaptive Stroke Calibration
i) Smart Punch Monitoring
j) Multi Tool Function
k) Cutting Packages for Material Types outlined in Table I

3.3.26.1 **Ethernet Connection Port.** An RJ-45 connection port for Ethernet shall be readily available on the exterior of the machine. The CNC controls shall have the capability to upload and download part program results using standard Ethernet protocol.

3.3.26.2 **Universal Serial Bus (USB) Connection Port.** A minimum of one (1) Standard USB connection port for the transfer of process information to the system control shall be provided. All protocol documentation shall be supplied.

3.3.26.3 **CNC and Control Restoration Data.** The contractor shall be responsible for the generation of all program logic and documentation necessary for machine operation. The Programmable Memory Controller (PMC) program, CNC parameters and all data required for configuring a new control and any other configured or programmed electronic device shall be provided in electronic format on read-only optical disc and delivered with the equipment. For safety and security reasons, the vendor shall not provide encrypted files containing proprietary cycles, PMC, PLC programs/data.

3.3.27 Motors. Motors shall be rated for continuous duty. Alternating-current (AC) motors shall be designed to operate on 60-Hz and each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of NEMA MG-1 or current equivalent ISO/EN/DIN/JIS/IEC standards. At the time of proposal, the vendor shall indicate which document they will utilize for compliance to the requirements. Each motor shall have an identification plate in accordance with paragraph 3.7 to identify manufacturer, model number, serial number, voltage, amperage, horsepower, phase, frequency, and frame number. Motor horsepower shall be the manufacturer's standard or standard optional horsepower as indicated on the manufacturer's design and construction drawing(s).

3.3.28 Facility Connections. Electrical connections made to the facility electrical system shall conform to NFPA 70 and be in accordance with Attachment C, Paragraph 3.1.8 (See Attachment D for source location).

3.3.29 Hour Meter. The machine shall be equipped with a non-resetting type, hour meter to display accumulated operating time of the drive motor(s), mounted on the main electrical panel visible from the outside, sealed to prevent the entrance of dust and moisture, and shall be mounted to withstand shock and vibration generated by the machine. The meter shall have a range of 0 to 99,999.9 hours in increments of one (1) tenth of one (1) hour. A CNC control that tracks and displays total machine on time (a non-resetting type), controller on, program runtime, and run time of the servo motor(s) shall be considered to meet this requirement. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.4 Size and Capacity. The size and capacity of the machine shall be not less than specified in Table I.

TABLE I	
Maximum Allowable Footprint (LxW)	44 Ft X 36 Ft
Maximum Allowable Height	16 Ft
PUNCHING	
Nominal Worksheet Size	120 inch X 60 inch (without workpiece move) 144-inch X 60 inch (with workpiece move)
Material Thickness	.016 - .250 in Thick
Material Types	Aluminum, Mild Steel, Inconel, Titanium, Copper, Brass, Nickel, and Stainless-Steel Alloys.
Max Work-piece Weight	300 lbs
Punching Capacity	30 Tons
Punch Turret Configuration	48 Stations with minimum of (3) 3.50-inch Auto Indexing Stations

Hit Rate	350 Hits Per Minute
Table	Brush and Roller Bearing combination
Part Chute Size	15 inch by 56 inch
Max Punching Diameter	3.0 inch
Programmable Work Holder Clamps	3
Traverse Positioning Speed	X,Y = 120 m/minute
LASER CUTTING	
Fiber Laser Power	4000 Watts
Cutting Gas	O_2, N_2
Power Stability	+/- 2% or Less
Laser Wavelength	1.08 µm
Material Types	Aluminum, Mild Steel, Inconel, Titanium, Copper, Brass, Nickel, and Stainless-Steel Alloys.
Nominal Worksheet Size	120-inch X 60 inch (without workpiece move) 144-inch X 60 inch (with workpiece move)
Material Thickness	.016 - .250in
Work Holder Clamps	3

3.5 Alignment and Accuracy Tolerances. The alignment and accuracy tolerances of the machine shall be as specified in Table II.

Table II	
Positioning Repeatability	±0.002 inches
Positioning Accuracy	±0.004 inches

3.6 Standard Tooling and Equipment. All standard tooling and equipment required for proper operation and maintenance of the machine shall be provided. A list of these items shall be furnished at time of proposal. The following shall also be included as a minimum:

a) Consumables for the initial operation of machine during origin and destination acceptance.
b) The contractor shall provide one set of spare consumable parts for their machine and cutting head for a one year period. A list of consumable parts needed shall be provided with the POAM.
c) A camera system for providing visual aid while obstructed laser cutting is being performed shall be provided. The camera system shall include a monitor attached conveniently to the control panel of the machine.

3.6.1 Required Tooling. The following tooling is required:

3.6.1.1 Stand Alone Tool Sharpening Unit. A stand-alone sharpening solution shall be provided with the following requirements.

a) Tool sharpening up to 3.0-inch Diameter
b) Grinding Motor Power 3 Horsepower
c) Wheel Rotation Speed 2750 RPMs
d) Table Motor Power of 2 Horsepower
e) Table Rotation Speed of 35 RPMs
f) Closed coolant system
g) Tool loading Chuck with chuck key
h) Intuitive touch screen
i) Auto Interlocking Safety Guard
j) Sharpening adapters that support the OEM machine working envelope (2)
k) Grinding Wheels that support the OEM machine working envelope (2)
l) Dressing Stone
m) Coolant for sharpening unit

3.6.1.2 **Required Punches.** For the first quantity of all tools outlined in Table III the vendor shall provide the full tool and holder to include the punch, die, and strippers. For any additional quantities called out the vendor shall provide the punch, die, and strippers only. All punch diameters identified shall include a quantity of one each die clearances of .006, .012, and .018in for each tool identified in Table III.

TABLE III

Type	Size (inch)	Quantity
Multi Punch Tool Holders for Indexing Stations		Three (3)
Turret Adapters for turret	B to D	Two (2)
Top Center Punch Set		One (1)
Bottom Center Punch Set		One (1)
Round Punch Dia. (inch)	0.06	Three (3)
	.062	Three (3)
	.072	Three (3)
	.076	Three (3)
	.078	Three (3)
	.091	Three (3)
	3/32	Three (3)
	.100	Three (3)
	.101	Three (3)
	.104	Three (3)
	.112	Three (3)
	.113	Three (3)
	.114	Three (3)
	.116	Three (3)
	.124	Three (3)

	.125	Three (3)
	.128	Three (3)
	.131	Three (3)
	9/64	Three (3)
	.149	Three (3)
	.150	Three (3)
	.165	Three (3)
	.170	Three (3)
	.172	Three (3)
	.173	Three (3)
	.174	Three (3)
	.175	Three (3)
	.176	Three (3)
	.177	Three (3)
	.178	Three (3)
	3/16	Three (3)
	.193	Three (3)
	.196	Three (3)
	.198	Three (3)
	.199	Three (3)
	.201	Three (3)
	.206	Three (3)
	.225	Three (3)
	.234	Three (3)
	.250	Three (3)
	.279	Three (3)
	.310	Three (3)
	.285	Three (3)
	11/32	Three (3)
	.355	Three (3)
	.375	Three (3)
	.388	Three (3)
	.404	Three (3)
	.500	Three (3)
	.687	Three (3)
	.750	Three (3)
	.980	Three (3)
	1.250	Two (2)
	1.311	Two (2)
	1.316	Two (2)
	2.500	Two (2)
	3.000	Two (2)
Square Punch (Inch)	1/8	Two (2)
	3/16	Two (2)
	¼	Two (2)

	5/16	Two (2)
	3/8	Two (2)
	7/16	Two (2)
	1/2	Two (2)
	5/8	Two (2)
	¾	Two (2)
	7/8	Two (2)
	1 inch	Two (2)
	2 inch	Two (2)
Rectangle Punch (Inch)	1.000 X .250	Three (3)
	3.000 X .250	Three (3)
Coining Style Countersink Punch/Die Set for .032 inch thick Aluminum sheet		
Top Hole CSK Size	.147 in	One (1)
Through Hole Size for CSK	.098 in	
Full Angle Between Sizes	90 degrees	
Top Hole CSK Size	.187 in	One (1)
Through Hole Size for CSK	.125in	
Full Angle Between Sizes	90 degrees	
Forming Tap Heads		Three (3)
Tapping Pitch Inserts	4-40 UNC 0.1in	Two (2)
	5-40 UNC 0.113in	Two (2)
	6-32 UNC 0.122in	Two (2)
	8-32 UNC 0.149in	Two (2)
	10-24 UNC 0.17in	Two (2)
	10-32 UNF 0.175in	Two (2)
	12-24 UNC 0.196in	Two (2)
	2-56 UNC 0.079in	Two (2)
	3-48 UNC 0.091in	Two (2)
	1/4-20 UNC 0.225in	Two (2)
	1/4-28 UNF 0.225in	Two (2)
	3/8-16 UNC 11/32in	Two (2)
	3/8-24 UNF 0.355in	Two (2)
	5/16-18 UNC 0.234in	Two (2)
Multi Tool 1 (Round)		Three (3) each
Set #1,8	.092 in	
Set #2,8	.098 in	
Set #3,8	.119 in	
Set #4,8	.124 in	

Set #5,8	.126 in	
Set #6,8	.185 in	
Set #7,8	.216 in	
Set #8,8	.249 in	
Multi Tool 2 (Round)		Three (3) each
Set #1,8	.093 in	
Set #2,8	.104 in	
Set #3,8	.109 in	
Set #4,8	.140 in	
Set #5,8	.156 in	
Set #6,8	.171 in	
Set #7,8	.177 in	
Set #8,8	.191 in	
Multi Tool 3 (Round)		Three (3) each
Set #1,8	.200 in	
Set #2,8	.219 in	
Set #3,8	.256 in	
Set #4,8	.264 in	
Set #5,8	.293 in	
Set #6,8	.326 in	
Set #7,8	.341 in	
Set #8,8	.372 in	
Roller Deburring	.125 in	One (1)

3.6.2 **Required Accessories.** The following accessories are required:

3.6.2.1 **Punch and Die Tooling Storage Cabinets.** A quantity of four (4) stationary tooling cabinets constructed to hold the punches and dies outlined in Table III shall be provided. The cabinets shall have eight (8) drawers and an overall dimension of 30-inch wide by 27-inch deep by 42-inches high shall be provided. The cabinets shall be constructed to withstand the weight of the tooling and be anchored to the floor.

3.6.2.2 **Computer Workstation.** Shall Include:

a. One (1) Vidmar or equivalent Type A-1 Straight workstation containing:
b. One (1) 1-SEP1023AL cabinet or equivalent, The cabinet shall be made of steel and powder coated Blue, having Five (5), 400lb capacity drawers and an overall dimension of 30-inch wide by 27-inch deep by 33-inches high.
c. One (1) BL1751 bench leg or equivalent, The leg shall be made of steel and powder coated Blue with a height of 33 inches.
d. One (1) HT60 hardwood top or equivalent measuring 60 inches by 30 inches with a thickness of 1 ¾ inches

3.7 **Marking on Instruments, Control Panels, and Plates.** All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion resistant.

3.7.1 **Lubrication Plate.** A lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information.

- Points of Lubricant Application
- Servicing Interval
- Type of Lubricant
- Viscosity
- Military or Federal Specification for Each Lubricant (If available)
- Size and Type of Each Lubricant Filter

3.7.2 **Nameplate.** A nameplate shall be permanently and securely attached to each machine. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records.

- Nomenclature
- CAGE Code
- Manufacturer’s Name
- Manufacturer’s Model Designation
- Manufacturer’s Serial Number
- Power Input (Volts, Total Amps, Phase, Frequency)
- Contract Number
- Date of Manufacture
- Full-load current for each supply
- Ampere rating of the largest motor or load
- Short-circuit current rating of the industrial control panel
- Max ampere rating of short-circuit and ground fault protective device, where provided
- Electrical diagram number(s) or number of the index to the electrical drawings

3.7.3 **Identification Marking of Military Property.** An Item Unique Identification Marking (IUID) shall be provided in accordance with MIL-STD-130N and all applicable documents within the standard with Machine Readable Information (MRI) for item identification marking and automatic data capture. The application of Human Readable Information (HRI) shall be used in combination with MRI and free text. At a minimum this tag shall contain the information listed in paragraph 3.7.2 Nameplate.

3.8 **Technical Data.** Two (2) hard copies and three (3) electronic copy(s) on compact disk (CD) of technical data shall be provided with the machine. Technical data shall contain specifications, instructions for the operation, maintenance of the machine and recommended spare parts list. All copies of technical data shall be legible and written in the English language.

a. All engineering drawings shall be in accordance with ASME Y14.100. Technical data shall contain technical information, specifications, instructions for the operation, maintenance of the machine, consumables and expendables information (such as oils, fluids, grease, filters…), and recommended spare parts list.

b. Factory service and operator manuals for the type and model of the machine being purchased. Manuals shall include electrical/electronic, pneumatic, and hydraulic schematics for the machine

c. Diagnostic procedures for isolating and correcting faults. Procedures shall be included in the service manuals for recalibrating and performing operational checks to ensure proper operation of any replaced parts, as required.

d. Preventative and predictive maintenance tasks, including oil and vibration analysis, and show recommended intervals of tasks.

e. Software manuals and registration keys/codes shall be provided, along with original software installation disks.

f. The Contractor shall provide the Government a copy of all test results completed in section 4.7 of this specification.

3.8.1 **Plan of Action and Milestones (POA&M).** The contractor shall provide a POA&M to identify contractor performance dates of all major events from contract award to completion of all contract requirements. The initial POA&M shall be delivered thirty (30) calendar days After Contract Award (ACA). The milestones shall be prepared using a recognized project management technique (preferably Microsoft Project). The Government and the contractor shall review the report on a monthly basis. The contractor shall notify the Government prior to making any changes to the POA&M. All changes to the POA&M shall be approved in advance by the Government. The POA&M shall detail such items as engineering, procurement of major hardware, assembly, operational debug, factory acceptance, shipment, installation, training, and final acceptance test.

3.8.2 **Acceptance Test Plan (ATP).** Contractor shall prepare an Acceptance Test Plan (ATP). The ATP shall be a step-by-step written instruction on how to verify compliance during Contract Quality Assurance Inspection and the Final Inspection/Acceptance Test specified in Sections 4.4 and 4.5. The Contractor shall develop and provide dimensioned drawings of the proposed test part specified in Paragraph 4.7.3. The Manufacturer's standard testing procedures shall be included in the ATP. Any procedures necessary for compliance with Section 4, which are not covered by the Manufacturer's standard testing procedures, shall be identified and included in the ATP. The initial ATP shall be submitted thirty (30) calendar days ACA for Government review and approval. The accepted test procedures shall become part of this contract.

3.8.2.1 **Part Model and Drawings.** The Contractor shall develop a test part model (DXF) for the performance testing outlined in section 4.7.3 The Contractor shall submit the test part model and drawing along with the POA&M. The test part shall utilize all laser, punching functions, and features of the machine. The Government will review the test part model within fifteen (15) business days of submission. The Contractor shall re-submit any required changes to the test part model within five (5) business days for review and approval by the Government. The Contractor shall provide the Government dimensioned drawings of the approved test part thirty (30) calendar days prior to Origin testing.

3.8.3 <u>Control of Hazardous Energy (Lockout/Tagout) Procedures.</u> A written procedure for isolation of all energy sources on all equipment provided under this contract, in accordance with 29 CFR 1910.147, shall be provided by the contractor and included in the technical manuals. The draft procedures shall be submitted in Microsoft Word compatible format thirty (30) calendar days ACA for Government review and approval.

3.8.4 <u>Equipment Drawings.</u> The Contractors proposal shall be provided with dimensioned equipment drawings in both a .PDF file and CAD file compatible with AutoCAD 2015 edition. The dimensioned CAD file shall be drawn to actual size and show footprint and any manufacturer's special foundation requirements of new equipment to include ancillary pieces, utility connection points, access doors and required maintenance clearance dimensions. The CAD file shall show side and elevation views.

3.8.4.1 <u>Installation/Foundation Drawings</u>. The drawings shall contain at least, but shall not be limited to, the following information and requirements of Attachment A:

- Overall and principal dimensions in sufficient detail to establish limits of space in all directions required for installation, operation and servicing. See Enclosure 1 for maximum allowable footprint
- Amount of clearance required to permit opening of doors and removal of plug-in units
- Clearance for travel or rotation of any moving parts
- Interface mounting and mating information, such as dimensions of location for attaching hardware, and meet the manufacturer vibration design requirement
- Interface pipe and cable attachments required for installation and co-functioning of the item to be installed with related items
- Mounting place details, hardware requirement lists, drilling plans and shock mounting buffers
- Weight of each component and total unit
- Location, type and dimensions of cable entrances, terminal boards, electrical connectors, and ducts
- Interconnecting and cable detail
- Peak electrical load (Amps) requirement to support all proposed equipment in its fully configured and operational state, including all accessories

3.8.4.2 <u>Installation/Foundation Drawings (Preliminary Submission)</u>. Not later than sixty (60) calendar days after effective date of contract, installation drawings shall be provided with soft metric measurements using the manufacturer's standard format. An electronic copy shall also be provided in DWG or DXF format in the receiving activity's version of AutoCAD. The Government shall notify the contractor in writing within 15 calendar days of receipt of drawing if they comply with this specification. If revisions are required, the contractor has 15 calendar days to make revisions and resubmit for a new review. Rejection of drawings does not relieve the contractor from final delivery date of contract performance period.

3.8.4.3 **Installation/Foundation (Final submission)**. Submission of final installation drawings shall be not later than sixty (60) calendar days of Government acceptance of preliminary drawing. These final drawings shall bear a complete title block with a permanent drawing number, signature, date, and an original stamp or seal of a Professional Engineer (PE). The final drawing submission shall include an engineering report with structural calculations used in the design. The engineering report shall also be stamped by a PE. The Government shall notify the contractor in writing within 15 calendar days of receipt of drawing if they comply with this specification. If revisions are required, the contractor has 15 calendar days to make revisions and resubmit for a new review. Rejection of drawings does not relieve the contractor from final delivery date of contract performance period.

3.8.5 **Instrument and Calibration Procedures.** Calibration procedures and/or tooling requiring operator verification or recalibration as part of a production run or on a daily or periodic basis shall be provided to the Government. The contractor shall provide a list of these items at the time of proposal. All items shall have a certificate of calibration traceable to the National Institute of Standards and Technology (NIST). The contractor shall submit draft procedures thirty (30) calendar days ACA for Government review and approval.

3.9 **Installation.** Contractor shall be responsible for the attached Installation Responsibilities Sheet and Local Documents Attachment A,B and C. The contractor shall follow the customers' hours of operation; 6:30am to 3:00pm, Monday through Friday excluding all national holidays, unless informed otherwise. Contractor shall be responsible for reading, understanding, and complying with Government facility requirements and information in Attachments A, B, and C.

3.9.1 **Rigging.** The contractor shall be responsible for all equipment (forklifts, man lifts, etc.) and qualified personnel for unloading, rigging, installation, anchoring, connecting machine to utilities, and leveling the machines in place. The contractor shall provide the necessary anchor bolts, leveling screws, and related hardware to secure the machines' components to a foundation.

3.9.2 **Field Supervisor.** Contractor shall provide a full-time field supervisor to direct installation and testing of the supplies from start of installation to final acceptance. The field supervisor shall be capable of reading, writing, and conversing fluently in the English language. The field supervisor shall have full authority to implement field decisions expeditiously. No work shall be accomplished when the field supervisor is not in the immediate work area. The field supervisor shall be identified to the receiving activity prior to start of installation. The field supervisor shall be responsible for coordinating with the receiving activity for any security clearances for all contractor personnel expected to perform on-site work. The receiving activity Government Point of Contact shall be notified of any change in personnel performing this duty.

3.9.3 **Installation Plan.** The Contractor shall be responsible for providing an Installation Plan in accordance with Attachment C, Paragraph 3.1.7. The Installation Plan shall contain detailed information regarding all installation work. The initial Installation Plan shall be delivered to the Government within thirty (30) calendar days After Contract Award (ACA) for review and approval.

3.9.4 **Compressed Air Connection.** The Contractor shall be responsible for design and installation of new compressed air piping and associated components in accordance with Attachment C, Paragraph 3.2.2 (See Attachment D for pipe size and source location). The existing compressed air pressure available at the Government facility fluctuates between 83 to 93 psi. The contractor shall assume that all compressed air qualities to be of the lowest quality that is useable as compressed air. The Contractor shall make provisions to filter, dry, humidify, lubricate, amplify, and regulate the Government supplied compressed air to meet the Original Equipment Manufacturer (OEM) compressed air requirements and recommendations.

3.9.5 **Machine Foundation.** The Contractor shall be responsible for providing a machine foundation conforming to Original Equipment Manufacturer (OEM) requirements and in accordance with Attachment C, Paragraph 3.2.4. For unforeseen issues that are found during installation of the foundation, the contractor shall notify the facility point of contact (POC) and the Contracting Officer immediately to determine further course of action.

3.9.6 **Telecommunications Connection.** The Contractor shall be responsible for coordinating with the FRCE Project Manager and IT group via the DLA contracting officer regarding communication signal connection of equipment to the manufacturing network along with any required stand-alone computer systems in accordance with Attachment C, Paragraph 3.2.5

3.9.7 **Geotechnical Survey.** The contractor shall perform a geotechnical survey to determine if and where there are any hidden utilities in the location of the current foundation. The onsite survey shall be completed within thirty (30) calendar days ACA, coordinate with the Contracting Officer.

3.9.8 **Lead and Asbestos Testing.** The Contractor shall be responsible for conducting an onsite survey for Lead Based Paint (LBP) and Asbestos Containing Material (ACM) in accordance with Attachment C, Paragraph 3.2.1. The onsite survey shall be completed within thirty (30) calendar days ACA, coordinate with the Contracting Officer.

3.9.9 **Work Stoppage.** For unforeseen issues that are found during installation, the contractor shall notify the facility point of contact (POC) and DLA immediately to determine further course of action.

3.10 **Training.** After satisfactory completion of acceptance testing of the system, the services of a qualified representative(s), who is proficient in the English language, shall be provided for specialized training to familiarize Government personnel with the equipment and to help ensure reliable performance and maximum service life, during normal usage. The training shall be in accordance with the following paragrahs 3.10.1-3.10.4. The course shall utilize as-built drawings and manuals provided under this contract and shall include both classroom and hands on portions. All printed training aids shall be supplied for the course, be in the English language, and become the property of the Government.

3.10.1 **Operator Training.** Operator training shall be provided at FRC East. Training shall be provided for a minimum of three (3) operators. Training shall not be less than three

(3) consecutive eight (8) hour workdays, between 6:30 am and 3:00 pm excluding weekends and federal holidays. Training shall include preparation of equipment for safe operation and proper isolation of equipment procedures. The operator shall be provided instruction for the care, use and operation of all attachments and accessories. Training manuals shall be provided for each student.

3.10.2 Laser/Punch Software Training. Following the completion of Operator Training, additional training shall be provided at FRC East or at an offsite training center for the cam software solution provided and called out on 3.3.21. The training shall be for a minimum of three (3) artisans. Training shall not be less than three (3) consecutive eight (8) hour workdays, between 6:30 am and 3:00 pm excluding weekends and federal holidays. The training shall include all functions and capabilities of the software. Training manuals shall be provided for each student.

3.10.3 Maintenance Training. Mechanical and Electronic/Electrical maintenance training shall be provided for four (4) maintenance personnel and two (2) equipment engineering personnel. Training shall be performed at the Government facility and shall not be less than three (3) consecutive eight (8) hour workdays each, between 6:30 am and 3:00 pm excluding weekends and federal holidays. Training manuals shall be provided for each student. The maintenance course shall cover at a minimum:

a) review of systems drawings and schematics
b) component location and function
c) troubleshooting procedures and techniques
d) repair procedures including disassembly and assembly
e) system maintenance and servicing
f) setup, adjustments, and calibration (how, when, and where)
g) preventative maintenance procedures
h) Procedures for proper isolation.
i) Basic start up, manual, auto, and MDI mode
j) SRAM, Ladder, PMC Memory, Parameters, Backup and Restoring
k) Troubleshooting PLC, Controls, and Drives using maintenance laptop and troubleshooting software identified in paragraph 3.3.22.4.

3.10.4 OEM Facility Maintenance Training. Training shall be provided for two (2) Mechanical and two (2) Electronic/Electrical maintenance personnel. The training location shall be performed at a certified training facility of the machines manufacturer and shall not be less than four (4) consecutive eight (8) hour workdays, thirty-two (32) hours. Training manuals shall be provided for each student. The maintenance course shall include but not limited to:

a. I/O Board Diagnostics and Addresses
b. Root Cause Analysis Troubleshooting
c. Machine Routine Repair, Replacement, Realignment Procedures

4.0 QUALITY ASSURANCE.

4.1 Responsibility for Inspection. The contractor shall be responsible for the performance of all inspection requirements specified herein and in accordance with the attached Quality Assurance Provision (QAP). Except as otherwise specified in the contract or purchase order, the contractor may use his own facility for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to witness any inspections set forth in the purchase description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance. The Contractor shall be responsible for ensuring all items meet the requirements of sections 3 and 4.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a) Contract quality assurance inspection (Origin) (see 4.4).

b) Final Inspection/Acceptance test (Destination) (see 4.5).

4.3 Inspection Conditions. All inspections, tests, and examinations shall be performed in an indoor facility with ambient temperature conditions in the range of 55° F to 100° F and 10% to 95% relative humidity.

4.4 Contract Quality Assurance Inspection (Origin). Contract quality assurance inspection shall be applied at the contractor's facility prior to being offered for acceptance under the contract. The machine and all peripheral equipment added to the machine shall be fully assembled and perform all functions in accordance with this specification. Quality conformance inspection shall consist of the examination in 4.6 and the tests in 4.7. The machine shall pass the examination, all tests, and contract quality assurance inspection to be acceptable for shipment.

4.5 Final Inspection/Acceptance Test (Destination). Final Inspection and acceptance test shall be performed on the machine to ensure conformance with this purchase description. The acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the examination and all tests to be accepted.

4.6 Examination. The equipment shall be examined for design, dimensions, construction, materials, components, electrical equipment and workmanship to determine compliance with the requirements of this specification.

4.7 Tests. Unless otherwise specified within this document, all programs, instruments, materials, material handling equipment, qualified personnel, and tools required to perform and evaluate the tests shall be furnished by the contractor. All measurement tools and indicating devices shall be calibrated in accordance with the attached QAP. If a malfunction occurs during the test, the test shall be restarted each time until all tests in 4.7 are completed without a malfunction. Three (3) consecutive failures without completion of the test may be considered cause for rejection of the system. For the purpose of this test, a "failure" is defined as any equipment malfunction which requires remedial action to restore the system to full operation in accordance with this specification.

4.7.1 **Alignment and Accuracy Tests.** Tests shall be conducted to ensure the machine meets the alignment and accuracy tolerances specified in paragraph 3.5, Table II. The machine repeatability and accuracy shall be checked.

4.7.2 **Operational Test.** The machine shall be operated under CNC at no-load for not less than four (4) hours. During operational testing proper operation of all control functions, machine response to commands, proper operation of motors, adjusting mechanisms, and auxiliary functions shall be checked. The machine shall be run in each mode of operation (Manual and Automatic) to test the machine's response to command data. Three (3) failures in a row to complete the run test may be cause for rejection of the entire system. After successful completion of the operational test, proper operation of all controls, motors, adjusting mechanisms, over-travel limits (programmed stops and hard stops), emergency stop devices, and accessories shall be verified. The safety limit stops shall be checked by attempting to exceed the limits of travel of each moving axis.

4.7.3 **Performance Test.** The machine shall be tested in accordance with ISO 9013 and within the parameters of the offered machines movement. Material types for testing shall include Aluminum, Stainless Steel, and Mild Steel. Material thickness shall be no less than .125 thick for all material types. The performance test shall include both punching and laser cutting operations. The machine shall also be tested in accordance with the agreed upon ATP referenced in paragraph 3.8.2 utilizing the test part referenced in paragraph 3.8.2.1. A minimum of three (3) parts per material type shall be produced and meet tolerances outlined in Table IV below.

TABLE IV	
Finished Workpiece Accuracy	±1/64 inches
Finished Workpiece Repeatability	±1/32 inches

4.7.4 **Audible Noise Level Test.** The equipment shall be tested for compliance with paragraph 3.1.4. "Audible Noise Level".

INSTALLATION RESPONSIBILITIES

Government	Contractor	
☐	☒	a. Provide machine foundation in accordance with the Manufacturer’s design requirements.
☐	☒	b. Furnish labor and material handling equipment for off-loading and placing item on foundation.
☐	☒	c. Provide and install anchor bolts and nuts.
☐	☒	d. Set and rough level the machine on its foundation
☐	☒	e. Level and align machine.
☐	☒	f. Connect machine to the provided utilities hook-up.
☐	☒	g. Provide all necessary tools, gages, and instrumentation necessary to perform the required tests.
☐	☒	h. Provide and charge all systems with fluids in accordance with manufacturer's instructions.
☐	☒	i. Material for performance testing at Government facility
☐	☒	j. Material for performance testing at Contractor's facility
☐	☒	k. Material handling, Rigging equipment and Qualified Personnel for performance testing.

GOVERNMENT RESPONSIBILITIES

☒ Provide utilities hook-up within 55 Linear feet of the machine.